ActInf Textbook Group ~ Cohort 5 ~ Session 17 (Chapter 7, Part 2) ~ 2/26/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 17 (Chapter 7, Part 2) ~ 2/26/2024 | 56:36

all right welcome back cohort
5 we're in our second discussion on
chapter
so is there anything anybody wants to
begin with can look at the text we can
look at any question we can add a new
question we could look at
pmdp
anything
um
anyone can write in the chat or raise
their hand or just go for
it yeah so I have a sort of more more
open-ended question I've been sort of
um thinking about you know doing a
little little modeling project
and um what I would like to se what what
I'm thinking about is that I'm just
wondering how I can use this active
inference Machinery to model some kind
of um kind of like a recognition task
like how would you approach doing
something like an amness classification
with um um active inference but what I
am interested in is is bringing this you
know one key difference um from the
active inference like um approach to um
the problem
that's you know basically completely
solved in the using artificial neural
networks but in a in a way that um you
know when you if you feed amnest to um
like a Transformer based classifier you
patchy it so it turn it into small
little chunks of image and you feed that

into a Transformer and the Transformer
kind of like sees the whole sequence and
then finally learns how to classify an
image I'm just sort of wondering um
thinking about like how to find some
sort of connection between between doing
something like that except getting the
neural network or some other kind of you
know um system to choose where to look
to to to sort of model the behavior to
be almost like like a sakad like I'm
seeing a part of an image and then I
have to make a decision where to look
next to
um you know
um resolve as much uncertainty as I can
about what what am I seeing and it
doesn't have to be something like an
amist it could be like a really really
simple images to start with you know
like 8 by8 pixels some generated I don't
know kns and Crosses or whatever but I I
don't really even know where to start
with that um it's a it's a very
open-ended big big thing but if you have
any any suggestions worth look I would
love uh some pointers yeah great great
questions well certainly what I'll show
now is not the end of the story but I
think it'll provide some motifs that you
can pick up
on here's a 2020 one paper with axle
constant so in this setting it was an
icade
Visual and one of their Novelties was
that they made a bounding box that was
kind of like the zone that was visible
in the Cade so the Cross Focus is the

center of vision and then the affordance
of the iate is to move to another cross
um and then that would move the bounding
box
um and it's a
cultural pattern recognition
task like based upon identifying these
higher order patterns of
Pottery so that's one visual way I think
this is a super fascinating thing
because in almost all visual recognition
tasks um either the whole image is
passed through if it's sufficiently
small like mnist or it's ch tokenized or
chunked or however um but here we bring
the action into Vision with actual
guided cognitive Vision moving to
uncertainty so that might be
um way more efficient in certain
settings um for example recognizing a
person like first there's like the
movement just empirically like there's
like movement to the body and then
there's a movement to the face and then
identifiable parts of the face that's
the pyade model and then on the mnist
the recent uh structure learning paper
actually specifically they they do
amnest um
and they
[Music]
um so the labeled data in amnest are the
true identity of the digit and then they
use their structure learning technique
to um develop the the Styles handwriting
styles of the digit so they're not doing
structure learning on how many digits
there are as as far as my reading but

they do look at different
styles like these are different styles
of the
digits so then it is learning a it's
doing it's saying how many styles of one
are
there and then it's finding it through
um structure learning which is the
proposal of a additional latent factor
and then basian model reduction pruning
the model back down for
redundancy
um they find strong performance on The
Mist the code is in mat lab and I've
looked into
it
with some language models you might be
able to bring it into something out of
Matlab or just isolate it into pseudo
code but the mat lb code itself it's a
little bit of
a
um it it it integrates a lot with the
broader SPM package so it's not just
like you can just pop out the the mat
lab script and run it because if you
follow the the the rabbit's Trail it
goes into a few other SPM scripts that I
don't think have been brought over into
pmdp yet but from a pseudo code
perspective it's very um
informative awesome that looks super
relevant thank you yeah but this one
does not have Cade this one the amness
digit is coming
in um
fully one interesting thing also on the
on the cultural

one
this um generative
model was actually this is before P
DP this generative model is the one that
we utilized for active INF for ants we
just said well what if instead of it was
an iade in a bounding box without
leaving a trace what if it was a
nestmate moving in a Zone leaving a
trace so that kind of opened up the the
like observing from afar moving your
eyes but not leaving Trace versus skin
in the game where the movement from here
to here is interpreted as a physical
movement cool great stuff thank
you this this paper AI I mean has other
benchmarks like the the recognition um
Sprite data
set maybe even a third
one towers of Hanoi classic TI 83
game there some very
interesting
shapes
cool any other thought or part on
seven why mdp right now is only in
discrete time so pretty much everything
that you encounter in Pim DP today will
be in the chapter 7 Spirit um
sanj's
python work as well as RX and fur and
Julia have continuous time
models okay can we go through the
matrices other than a to d so you're
talking about in here um Pros or in
which figure do you um uh maybe yeah
this this seems good sure
okay
okay okay so here's the progression of

how they introduce the the matrices in
seven o is the observation Vector at
each time point so if you just had one
binary sensor o would be a Zer or one if
you had eight floats sensors it would be
eight float numbers s is a vector
carried Through Time Each time point of
the latent States so let's say we're
talking about temperature in the room
thermometer observations latent
temperature hidden State s agents
estimate of hidden State S A is a matrix
that has um rows and columns
corresponding to O and S A is a mapping
between o and s and so it can be used to
start with s and then emit a
distribution of expected observations or
it could be taking in an observation
conditioning upon an observation like
Bas theorem does and then updating um a
hidden State belief D has the same
dimensionality as s being carried
through time so D is just like a vector
it's kind of like t0 it just starts the
chain going and it gives you s at the
first time point and then B is a
transition Matrix that has edges and
rows and columns edges of the
dimensionality of s because it Maps s at
tus1 to S at T so this is a standard
hidden marov model ambiguity Matrix a
between hidden States and observations
transition Matrix B between Ates at
times there's no action in this model
yet um the next thing that they do is
they take that exact downstairs Model E
coming
down and then this is where the action

enters the picture which is pi which is
a the policy space expressed as a
probability distribution over actions so
Pi sums up to one it could be sampled
from or it could be just pick the
highest number or it could be you know
pick the second highest or all these
different things that you can do
regarding how to pick a policy from PI
but Pi intervenes with a policy in B and
so while previously B was just a single
Matrix going from s of T minus one to S
of T now B is going to be having a
matrix slice for every single affordance
that we have so each slice of B is like
an index card that describes the world
transition model conditioned upon action
selection G is expected free energy and
G is a distribution that essentially
updates the policy prior habit with
respect to epistemic and pragmatic value
equation
2.6 so that gets us to this kind of
figure 4.3 or figure
7.3 discreet time classic Model we have
downstairs hidden Markov model with no
action and then
actions select which B Matrix is going
to be
applied as s unrolls through time and
then um in seven figure
710 we see a lowercase letter above each
of these other letters that we've talked
about
that represents um a hyper prior or
however you want to think about it but
it's a distribution which that uppercase
variable is drawn from so instead of

just saying well um there's four states
in the world so r d Vector could be 0.
2525 2525 that' be like a uniform prior
across initial location or it could be
one z00
Z just unambiguously starting in the
first distribution or you could
parameterize Little D which would be a
distribution which D Big D is sampled
from for example across individuals or
across nestmates or across trials so
then learning could happen on Little
D across trials so you could begin with
like a uniform prior of where I begin
and then with learning by counting you
could just say okay I'm going to um with
the conjugate prior the de lay
distribution
I'm going to add one count to Little D
every time I observe myself starting in
a certain location and then trial after
trial you'll end up drawing Big D from a
prior distribution that does resemble
the experienced history that's the
learning by counting part and then the
end of the chapter they just kind of
present one more Motif which is now that
exact 7.3 three or figure 4.3 exact
Motif they're just demonstrating a
little bit of the composability of the
base graph by showing that like the
observation being emitted from a top
level model here level two that
observation could be understood as for
example a prior on a lower level
model but figure 4.3 or figure 7.3 is
kind of where in the descript every
time we get the

downstairs sense making part sense
making from
observations transition World model and
then finally to the action dependent
World transition
model yeah thank you it's I think it's a
very good overview and uh I'll refer it
back again but I think it's I just have
one question when you use the word
motive all the time what does it exactly
mean like I think I I just don't
understand that word actually sorry
Motif I use yeah what do you mean by
Motif sorry I I use it to like a pattern
like a pattern language or like an
identifiable theme so um like let's just
say we were looking at cars as
generative models like one of them is
red another one has convertible top
another one has six wheels another one
has a trailer like those are just kind
of like
motifs some motifs might be mutually
incompatible but other motifs it might
just be like well
like um here we're looking at the motif
of having a hyper
prior here we're looking at the motif of
nesting now the lower the lowercase
letters are not shown but of course you
could have lowercase
letters so there's nothing contradictory
about those two motifs it just and this
is definitely not like the only or
formal way to think about it it just
like here in figure
7.3 we kind of have like the kernel
nucleus um there isn't much you could

remove here if you wanted to remove
something you could remove like this
part so it was just focus on the
transition of two time
steps but that's very minor um this kind
of shows the basic
initialization and then the step from
the previous moment and then the step to
the next moment of
action as reweighted by expected free
energy so here we have kind of an
essential kernel and then there's all
these motifs or there's other ways
probably to say it but there's all these
other strategies that we could start to
see or
use Sor right thank
you and and
um yeah I mean I don't know how far the
uh Linux kernel metaphor will will go
but like this model doesn't have just as
written various of the more
sophisticated cognitive
phenomena so it's not the advanced
cognitive phenomena or higher order
human type thinking um is already in
this
image it's like this is the
canvas that is a starting
point for like learning a
novelty so it's not not that all these
different diverse cognitive phenomena
are like intrinsically essentially
within the active inference formalism
per set it's that we have something
that's General and modular enough to
accept and even entertain multiple
different Renditions of a given

phenomena like there's not just
learning a narrow definition of learning
would be like each parameter being
updated and then if somebody wanted to
engage with a higher order like kind of
conversationally what we might talk of
as
learning like chapter 6 is there for us
we could go through it and we would find
like a room of 10 people could come up
with a thousand ways to implement
learning at a higher level in active
inference and
a lot of emphasis on like what's the
core and then what are some of those
like decorators and modifications
augmentations and so
on because chapter 6 Plus The Core
Essence plus the confidence to play with
motifs gets very
far um would people be up for maybe a
little if we take a look at figure
7.10 would people may be up for trying
to walk through it perhaps like and I
don't know if this is applicable it's a
long way from the teamas but like
walking through it like with like a real
life kind of decision idea like what I'm
thinking is and try to map that process
onto the
model so like what I was thinking it's
like I'm sitting in my room and I'm
looking outside of my window and I'm
trying to
decide whether I'm going to go outside
or not and like looking at okay what are
the parameters of that decision what are
the different

um you know observable States or part
like or hidden states that are going
to influence you know that that
policy is you think that's an
appropriate exercise Daniel or something
along those lines would people be into
that I mean it it definitely is I'm just
finding a table that I'll copy
in
okay I'll make this under chapter
okay this is just one little
template so here we have the ab B CDE e
and pi and O and S so we um have we
don't have the the the lower case
letters yet but we could add them and um
each of those discrete time
variables is linked up to an ontology
term most of which which are all of
which have been used um and then here is
an example like for an ant calling
estimate how those got mapped um but I
mean let's hi hide these and then let's
just go for
okay what's the
setting what is the
agents living room living room in New
Mexico okay if that makes sense
yeah aren't we all okay
observing the the window out the outside
through the window so I guess outdoor
Outdoors through through the lens of
window okay pixels coming from outside
sure and then what's truly out
there could be is it raining or not yeah
the weather I guess you know is it is it
that's the decision I'm trying to make
yeah
okay well and and and this just goes to

show like first off how open it it like
what actions are but um I
mean we'll get there but can we control
the
rain so then there's already such
differences between a model where like
there's a hidden state that you can't
control but you still could make a
policy decision to like take an umbrella
or Not Right But then there would be
like another level of kind of
interactivity of a model when the policy
can also control the hidden State like
when there like a like an air
conditioner with a temperature versus
just sort of like a put on a jacket if
it's raining okay so now given that we
already said SN
o what is the dimensionality of a and
what is the semantics of
a let's just say just just to be clear
that the the visual stimuli that we're
getting is 10
pixels and then they have a continuous
level of brightness like we could go
into the RGB and all this other stuff
but there's 10 pixels and then raining
or not is a binary state
so what is the dimensionality of a or
what does a
map I mean not the the site of raindrops
are not right so like it
yes it it it it Maps but that's two
dimensional
yeah Maps between the observations in
the hidden state so its shape is 2 by1
Matrix okay I see
yeah there there's other we're just

going with just the first sketch pass
okay so this is mapping the these pixels
from the pixels down to beliefs about
the rain or vice versa so if we said
condition upon I know it's raining we
could emit a profile of pixels that's
the likeliest thing or we could take in
an empirical pixel distribution and then
ask what is our conditional probability
of it being
raining so that this is and then last
piece of the
downstairs sense making part what is the
B Matrix
shape given it's a transition
Matrix should be 2 by two I guess
yes these are just the transition
probabilities of staying
raining staying not raining or
switching the the on the
identity Matrix the diagonal on B is
like stay because you're mapping from
let's just say that the first um row is
raining the second one is not raining
the upper left corner on B is rain to
rain and then the lower right corner on
B is not rain to not rain and then the
off diagonals are
switching
okay this is the
downstairs part this is a partially
observable marov decision or sorry this
is just a partially observable marov
model no decision has come yet right now
we're going to come into the decision
okay so what is the action
space go outside or
not okay now if we go outside then we

we're going to get different
observations but we'll just say yes go
outside or not
okay um what is what are the
affordances basically not
raining it's an affordance for us to go
outside yeah um almost here the
affordances are the same as the policy
space because there's no planning okay
so affordances are what you can do in a
moment like up down left right and then
the policy space is the forance space
exponentiated to the time Horizon of
Planet so if it's a non-planning model
then affordances equals the policy space
but if it was um a two-step model then
you could plan I'm going to not go
outside and then I'm going to go outside
or I'm like I'm going to go outside and
then I'm going to come back so then that
is considering over multiple time steps
but here
single single time step no
blame okay um oh we need d as part of
our downstairs D
is prior on it
raining and so this is just this has the
same shape as
s so this basically is just this could
be like one zero which is
like we start sure it is
raining or it could be 0. 5.5
we start unsure if it's
raining okay and then um
actions are selected different ways
obviously that's what we're interested
in so what are some different ways you
could select actions well you could just

draw from your habit
distribution that's one approach in
um okay now we're going into the the the
New Mexico dopamine
brain on the left
side this is where we have actions being
drawn from The Habit
distribution so that's a totally totally
um viable decision- making strategy
that's kind of like thinking fast like
type one is drawing from
habit
um another approach is to use expected
free energy to re weight the policy
prior into a policy
posterior according to equation 2.6
which is to say reweighting policies
according to their epistemic plus
pragmatic
value so in this setting the epistemic
value
um right now like everything's pretty
much
fixed so there really isn't much
epistemic
value to gain
yet pragmatic value though is going to
be about preference
satisfaction so what would the
preference
be preferen is over
observations all I'll just note like
this recent one on intentional
Behavior
that they do explore like essentially
preferences over whether it's raining or
not but suffice to say in the textbook
and in this kind of primary version the

preferences are over
observations so let's just say that um
let's just say that it's um darker when
raining
lighter when
not so if our kind of if our
conversational preference is like I
prefer it to rain our C preference
variable because it's over observations
would say I I prefer I expect and prefer
it to be
darker because the preferences are
grounded in sensory
observations that haven't come in
yet
I mean uploading the textbook to a
language model or not copying in this
table and then saying write active
inference python code given that this is
my
PDP you will get a
script with just this much
information because you've given all of
the information
needed there's there's no other there's
no
there's way more complex models to make
but in terms of describing all of this I
mean G is not really a variable it's a
functional that gets applied to Pi
conditioned upon c and e so by defining
a b CDE e s
o and then the generative
process
the true
rain but that doesn't have to be active
inference that could be um a random
number that could be an API to a weather

station that that it could just be
anything else that plugs in and outputs
observations and then intakes actions
but here it's kind of a a null action
intake because you're choice to go
outside does not
affect yeah that makes
sense all right yeah thank thank you for
that cool yeah I mean plug it in and see
where it goes or um look at a
um documentation and see which one it
looks most
like I think we looked at it a previous
time but this one is a moving grid World
agent in
one and then
two
um Pros that this is
a under chapter
7 yeah um and then chapter
two here is
is again similar and then the
uh I
mean make some kind of Fusion variants
like there's a t- maze it's like your
living room is the starting position in
Maze and then like you know you could
check the weather that's the epistemic Q
or you could choose to bring an umbrella
or
not and then it would already have
almost isomorphic relationship with it
with a
tutorial
and especially when when starting this
is like why the iterating on having a
model is so informative like ironically
or or appropriately depending on how you

see it's like it's epistemic chaining
the modeler is in an epistemic chain
with their own
modeling it's super easy to wait on the
sideline until what I really don't know
but by putting something down then
there's all these other things that open
up so that's the epistemic chaining as a
um okay Francis let me just see what you
wrote is it straightforward to download
either the PMD the PDP notebook is the
chapter 7 document also okay
um about
PDP yes at the top of each page like
I'll just put it in the
chat you can open it in collab super
easy and then just make a copy so those
are very nice notebooks to pull um is
the chapter 7 document also a Jupiter
notebook
no um that would probably
require going to appendix
C and uh going to the mat lab
code or um this the team is
foraging and bringing this Matlab code
into something like a notebook I'm not
actually sure if mat lab even has
notebook or realistically since it's the
teamas
like this pmdp Taz
demo it's the same exact
thing so this notebook is probably a
good one to go
with uh I don't have collab uh as a
non-academic it's
expensive uh or used to be expensive is
it um sorry you said you don't have Koda
or mat lab

collab Co it's free you just click on
the link and then just duplicate
it okay right so
okay that lab does have a costly license
except there's the octave like variant
that's open source but like rather than
get into this whole like octave mat lab
situation
like the the python and Julia are the
way to
go and rust when the time is
right
this is
interesting recommend breaking the demo
to change generative model only by doing
this will an intuitive sense mechanics
of active inference
develop pretty interesting that's
exactly what I was exactly what I was
planning to
do thank you okay yeah I mean it's it's
super fun and then also so like again
seeing the kernel execute like just
seeing a decision be
made that helps open up like oh there's
so there's like all these levels of
Diagnostics or analytics about the trace
of the model that you might be
interested in now is that active
inference well it's about an active
inference model but then a Next Step
might be like to do a parameter sweep
across parameter combination and then
summarize the outcomes of those swept
parameters with another heat
map and then what do you do from there
you know then you could fold that back
into the base model or maybe you make

another active inference agent that
looks at the heat map and then does
something from
there hence it being more like lines or
journeys of
inquiry rather than like developing an
artifact that just kind of slams the
door
especially with these kind of more
intuition
pump
examples
um without wishing to derail things uh I
have an interest in doing uh test
automation for code and I was looking in
um chapter 7 at the
uh way that the agent uh explores
little maze quite efficiently because
it's
seeking
novelty um and I was wondering whether I
could do something like that to um do
exploratory testing of say an
API
um that that sounds very interesting
I've never specifically seen it but
something like adaptive fuzzing or
testing
and then that's where the epistemic
value comes into play like let's say
pragmatic value is I prefer the program
to work so and then epistemic value is
I'm going to find out about where and
how it does or doesn't work so if you
only um made the breaking changes that
you knew were going to break it it would
be very confirmatory but it wouldn't be
epistemic because you would only have

been confirming which which might be
valid exercise but you'd only be
confirming the consequences of the
actions that you already suspected but
maybe that's important and then another
setting would be like with a kind of
uniform prior or with an informed prior
what kinds of changes would actually
reveal more learning about the structure
of the program or the API and then it's
like but once we figured out that that's
a breaking change like we've kind of
mapped that part of the
maze there's probably a lot of cool ways
to go
there
Kristoff so I think it's it's an
interesting problem what to do with the
something like you know exploring an API
um there's this paper um
for planning uh no certain that planning
navigation
um uh active inference and int in
intentional behavior and they have this
um uh notion of um uh sort of like
inductive planning where there is a
certain goal that the agent wants to um
uh reach and it works backwards to
figure out what are the steps to get
there uh for something like you know
exploring API and I think a lot of for a
lot of applications though I think
what's going to be eventually necessary
somehow is to have a hybrid model that
you will be you will have to be able to
attach almost like an llm to it to
basically at some point say like well I
would like to actually you know I

decided on my course of action I want to
write some code that actually invokes
the API and here with this decision um
the structure of decision- making we're
looking kind of like at assembly level
of intelligence and at some point just
like we are attached to an to an llm
that continuously pumps out
words um and I I'm just you know sort of
choosing over over the distribution of
those words they're being pumped out um
but I'm not actually generating them
right like I'm not aware of how that's
happening um in the same way you would
have to kind of attach an llm to almost
like an active inference Ang that does
some kind of goal oriented exploration
of the API so that would be very
interesting if you could achieve
something like that it's probably you
know um quite a quite a lot of money to
be made on on on this sort of
intelligent exploration of apis given
how how much um uh trouble bugs are
causing cool thanks yeah like and then
in it kind of like in in that um like we
were talking about right before began
with like the feedback and the training
examples like you can have a data set of
correct API calls and incorrect API
calls or something like that um and then
like plan to make it
work and those are kind of the semantics
of perception and
action that like ecosystems of shared
intelligence synthetic intelligence if
we can construct a formalism that has
the semantics that are how we think

about it Andor co-evolve that with
updating how we think about
it then we would have kind of a high
bandwidth or a low
loss or some other good property of the
edge between us and the constructed
artifacts yeah I think this is where you
know like lm's really kind of suckut
that you can you can sort of ask it to
do something like this but it will not
really have any kind of notion of
thinking about the problem deeply about
how would you go about breaking uh
something for an API because that
requires a sophisticated model of what
the API might be doing under the hood
for instance like some kind of buffer
overflow and something like that but if
you could if you could come up with a
minimal example that would Muer that
would model something like you know
assume there is a possibility of a
buffer overflow in an API how would you
go about making some kind of inference
about um or planning to exploit that
that would be super interesting as a
minimum as as a kind of a minimum
example even for just a simple you know
very simple IPI that takes like a a car
star in in a size or something some
something like this um how would you go
about you know inferring whether it's
possible to to do the buffer overflow in
this one one funny um piece there would
be use check out the PDP API
and then the what would be appropriate
the appropriate um the
latitude within that API is active

inference that's kind of one that would
be a plot twist and then also on the
interpretability here's um a paper with
MAO and others from last year um like
largely there aren't technical advances
in this paper the images are drawn from
saned Smith's live stream 28 paper
so it's not like this is breaking new
theoretical ground but the whole point
is when you ask a neural network or llm
Etc you know how sure are you about that
it can come up with a sequence of
tokens but that sequence of tokens is
only going to be coincidentally related
to anything like an uncertainty
evaluation whereas if uncertainty
parameters are explicitly
encoded then
self access
introspection is just like a parameter
lookup it's not like some second it's
like part of the real time function of
the system is um uncertainty estimates
of different kinds at different levels
so then to ask about how certain it is
about a certain thing at a certain level
you just ask that question and then
return that
variable um you know maybe even that's
on some kind of special chip or
something like with um verified access
or something so it opens up to all these
different
architectures rather than like well
let's train like another adversarial
neural network to like you know try to
infer this about this one and then that
gets into this whole like deception game

versus having these introspect
architectures
yours to what extent does this differ
from basan reinforcement
learning
great what what what about that or what
what you you mean overall house active
is inference different than
reinforcement learning or yeah yeah
maybe indeed even overall but uh yeah if
if you want this kind of uncertainty
measure with whatever you infer then I I
guess both Frameworks will allow this or
is there yeah
uh is it's probably very unclear
question but yeah okay there's there's
there's a few a few works and a few
approaches there so
um models that aren't active inference
still do deal in uncertainty so there
still is an uncertainty parameterization
in even a linear aggression however a
neural network or reinforcement learner
that's trained to generate
language its outputs of languages don't
necessarily represent a veritical
introspection maybe there's a way to
train it so that the output is a
vertical representation but that would
be a secondary kind of an U welding
together whereas this architecture that
we just looked at like it's it's a
trally part that's kind of on the the
uncertainty as part of the architecture
part um the key the key difference
between active inference and the
reinforcement learning is in reinfor
ment learning a reward distribution is

proposed secondary or auxilary or cor
corollary distribution is proposed
that's like layered over the
observations so in the homeostatic
environment we'd say we're getting
observations about temperature and we
are rewarded by 37c and we seek out High
reward in active inference we do away
with this secondary auxilary function
and we directly work on on the
observation distribution by calling our
preferences not a preference for higher
score on this proposed distribution but
rather we have an expectation and a
preference that connects directly to the
observations
themselves
um yes I I understand that but I have
been
uh vaguely looking at uh um things that
Benjamin V Roy and uh his lab are doing
and but I I don't really understand it
yet but that's there is some kind of
thing like beijan reinforcement learning
uh and I uh I wonder if they have I I
would expect it to have a um a measure
of uncertainty that comes with with any
prediction there but I might have uh
yeah it might just be a too vague
Concept in my mind it no you're right
there it is there I mean this work on
um demystified and compared and there's
several
other works that do directly tackle
it it it's not out of the picture that I
mean architectures can include arbitrary
components so it is possible to be like
well let's have an epistemic module well

yes but then epistemic and pragmatic
value at the end of the day in the RL
learner or the Deep Q learner or
whatever epistemic and pragmatic value
are going to get Blended back
together into the reward function right
you have these like artificial curiosity
and and other uh intrinsic motivations
that you could add add on to a classic
canonical reinforcement learner but and
I see quite a lot of
of research on that but then it it seems
that from an active inference framework
it it kind of emerges from the from from
the the system but
um absolutely that is the Hope like that
that there's a variety of cognitive
phenomena from planning to epistemic
Value curiosity intentionality variety
of phenomena that could be implemented
in a in a range of
architectures but will pursuing a first
principles approach lead to simpler more
interpretable more efficacious models
possibly possibly um if if somebody
needed uh uh you know there might there
might be a better model for a situation
today because there are such incredible
gyms and toolkits for reinforcement
learning um so but but
yes hence the moment of the
epistemic excitement that we have and
share here plus the accelerating
conversion and understanding like yeah
where is this
useful because it's it's all good for
something to be
insightful but then um I mean there

could be a situation where active
inference just gives us an intuition
pump and then we use a neural network or
there are hybrid architectures that that
do this or that and um but if utility is
is what is
desired or or some other property that
another system already has then yeah
definitely active inference has to work
to to to be superior there one really
interesting um moment
from um this was a couple days ago
with uh Ryan Smith and two of his PhD
students so the empirical status of
predictive coding um what was super
interesting about this paper was
like in this textbook group if someone
said like what's the status like of of f
or status of ACM people would often take
it like in a metaphysical
Direction however they um took it the
other way and asked what's the empirical
status of predictive coding and active
inference in explaining
neuroimaging and
neurobiology
um so the it's very interesting and
these are these are the ongoing
questions and curating the models and
being able to compare them based upon
how they do treat these
phenomena that would be super
informative and then if we can have a
similar
um Syntax for the variable similar API
use agent I mean this
um the demystified and compared it uses
the open AI

gy so
like they presented some promising
interesting results and that was several
years ago and a lot of the work has
probably not been published on the open
AI octave inference side but it is right
there like with frozen lake or with um
the other reinforcement learning
settings then look at where they're
different find out where does the super
high performance large scale RL model do
well that this simple act imp one
doesn't and then what would we need to
do to make it
different well fun yeah chapter 7
is pretty
fun um next next weeks we'll head into
chapter eight on the continuous time
maybe we'll look at some RX and fur
examples because there aren't going to
be pmdp examples for chapter
8 um The Continuous time models there's
there's a lot to say there on the math
and on the philosophy and so
on chapter nine integrating with data
and then chapter 10 just a long wrap up
with no figures or equations
so thank you all see you next time thank
you Daniel
bye everybody